SC4001: Neural Networks and Deep Learning

Programming Assignment

Part A: Regression with DNN

Predict Singapore **housing price** from both **categorical** and **continuous** inputs

Question A1 (baseline model)

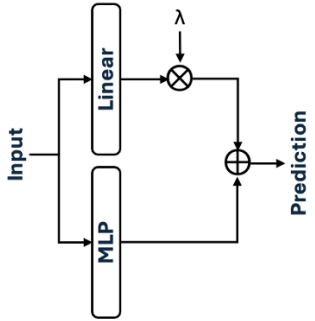
- Each categorical feature is first indexed and then converted to an embedding of dimension 8 by using **nn.Embedding** layer consisting of learnable weights.
- Draw **a block diagram** of the baseline network; compute the number of learnable parameters
- Find the optimal parameters: **embedding dimension** x **hidden-layer width** by using **ray[tune]**
- Use these parameters for A2 – A4

Question A2 (ablation)

- (i) Remove categorical variables

  (ii) Replace ReLU layer with sigmoidal layer
- Perform experiments on **five seeds** (torch.manual_seed(seed)), report mean and s.d., and compare with the baseline

Part A



Question A3 (wide-and-deep architecture)

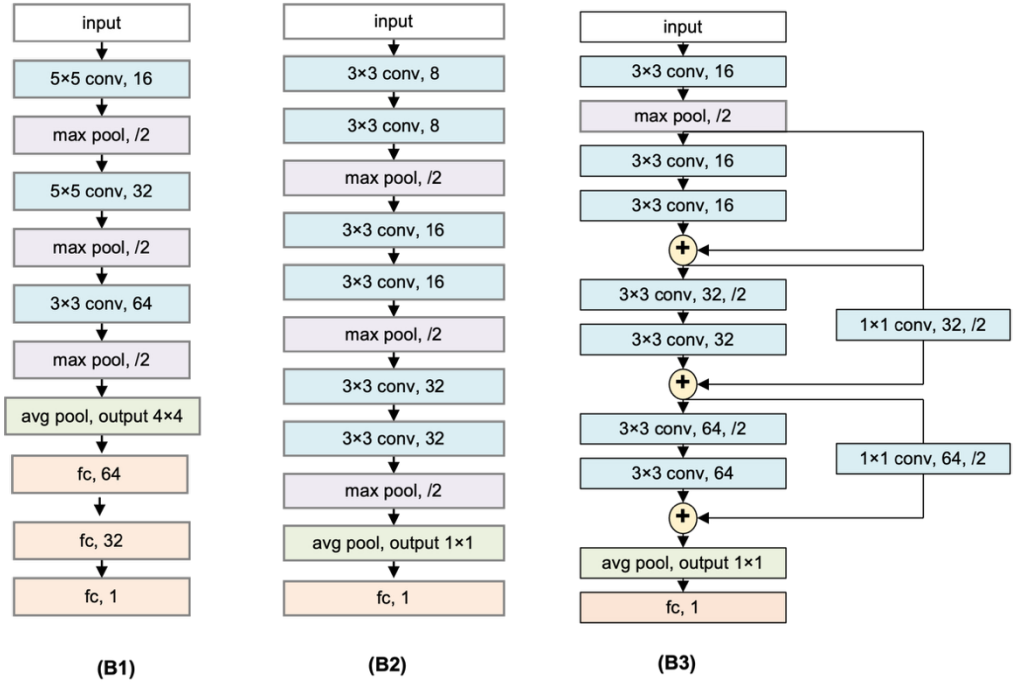
- output = MLP output + $\lambda \times$ linear output

  Find the optimal parameter $\lambda \in \{0, 0.25, 0.5, 1.2\}$
- Evaluate the optimal network over 5 seeds and compare with the baseline architecture (MLP)

Question A4 (XAI)

- Use **FeatureAblation** method on **Captum** library
- Find feature **importance scores** in Singapore dollars

Part B: classification with CNN



(B1) ***AlexNet*** *like architecture, (B2)* ***VGG*** *like architecture, and (B3).* ***ResNet*** *like architecture.*

All the layers use ***ReLU*** *activation function including after residual addition ⊕. All convolutional layers use* ***BatchNorm*** *and* ***SAME padding****. Layers indicating /2 have a* ***stride*** *of two and all other layers have stride of 1.* ***Dropout*** *= 0.5 is applied only in the hidden fully-connected (fc) layers in B1 architecture.*

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Part B

Question B1 (AlexNet-like architecture)

- Find the **optimal learning rate** from $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$
- Plot learning curves, evaluate, etc.

Question B2 (VGG-like architecture)

- Find the **batch size** from $\{16, 32, 64, 128, 258\}$
- Plot learning curves, evaluate, etc.

Question B3 (ResNet-like architecture)

- Use learning rate from B1 and batch size from B2; Plot learning curves, evaluate, etc.
- Compare B1-B3 architectures: performances, number of learnable parameters, FLOPs, etc.

Question B4 (XAI)

- Use **Grad-CAM** method on **Captum** library
- **Overlay** Grad-CAM heat maps on X-ray images

Housekeeping rules

- Assessment is based on **the report** (**6 pages each**). Codes will be used for checking.
  PartA_report_YourName.pdf, PartA_codes_YourName.zip
  PartB_report_YourName.pdf, PartB_codes_YourName.zip
- **Codes** must be in .py and .ipynb, tidied, properly labelled, and ready to run. Include a **README** file.
- **Generative AI declaration**
- Marks: **45** for Part A + **45** for Part B + **10** for **presentation**. Follow the instructions closely. Any deviations may lead to loss of marks for presentation.
- Late submissions: **penalized** for 5 marks for each day up to 3 days.
- This assignment is to be done **individually**. Absolutely NO copying, duplicating, or plagiarism.
- Post your queries on the **Discussion Board** in NTULearn
- Specific questions for help via cx4042@e.ntu.edu.sg
- TA: Deepank Girish, Yan Yige, Tiara Natasha, and Ajith Senthisenan.